AGRIVIVE / POKKALI-VENT

KERALA / POKKALI-INSPIRED / ADJUSTABLE DRAINAGE RISER

SERIES 12 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



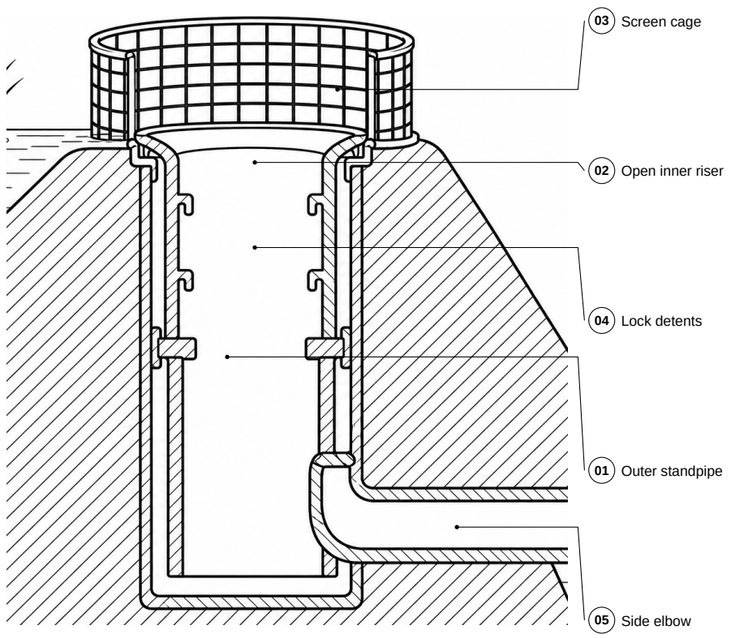
STATE CONTEXT / INDIGENOUS INSPIRATION

Pokkali landscapes in Kerala combine seasonal rice cultivation and prawn farming with managed water exchange. POKKALI-VENT is a new small gravity-drainage riser inspired by level control; it does not remove salinity.

Practice source: Kerala Industries Pokkali Rice GI profile | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 50 standpipe OD x 240 H

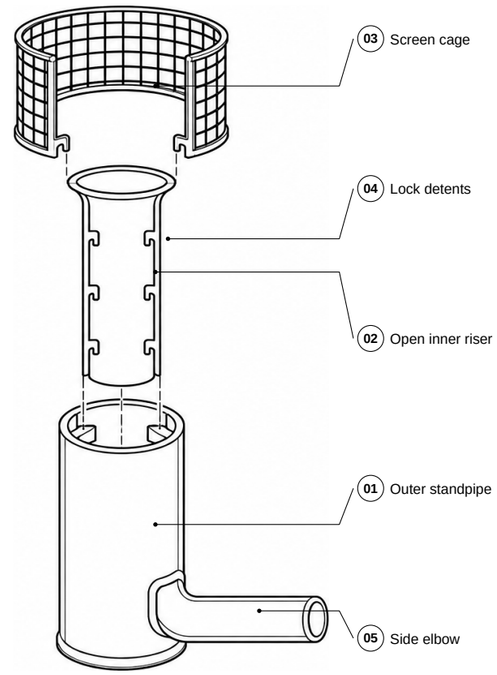


DESIGN / OPERATION NOTES

Seat pipe with outlet invert below the open overflow lip.
Adjust inner riser and engage the twist-lock detents.
Fit the coarse cage; water spills over the open lip and drains by gravity.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
50 standpipe OD x 240 H

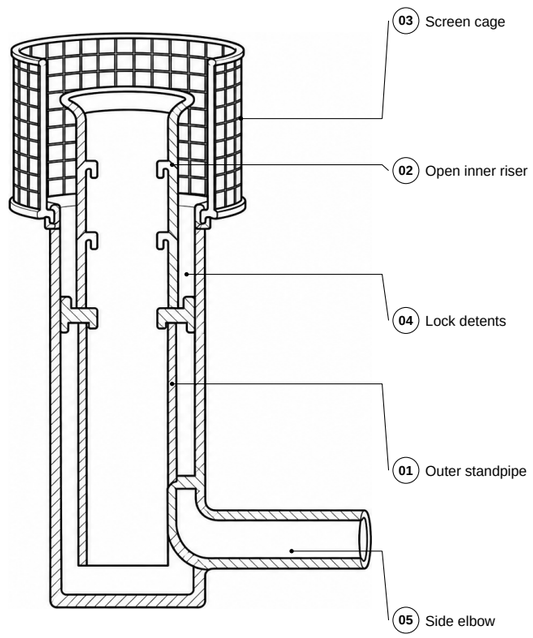


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
40 clear bore ID; 30 discharge ID; 60 candidate stroke



DESIGN / OPERATION NOTES

Adjust inner riser and engage the twist-lock detents.
Fit the coarse cage; water spills over the open lip and drains by gravity.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Standpipe OD / nominal H	**50 / 240**
Clear bore / outlet ID	**40 / 30**
Candidate riser stroke	**60**

02 GEOMETRIC CHECK

DISCHARGE BORE AREA

706.9 mm2

pi/4 x 30^2
No salinity-removal claim; needs gravity head

03 PARTS / QTY

01 Outer standpipe	**1**
02 Open inner riser	**1**
03 Screen cage	**1**
04 Lock detents	**Integral**
05 Side elbow	**Integral**

Pipe/riser/cage: candidate polymer. Seal and debris-screen aperture TBD.

04 INTERFACES / CHECKS

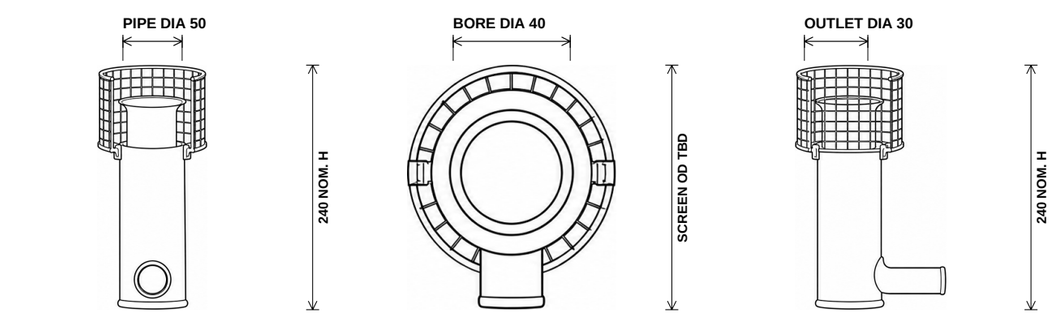
Check gravity fall, blockage and downstream backwater.

Test riser locking, sealing and withdrawal access.

Confirm screen size and assembled height at each setting.

PRELIMINARY / VALIDATION PENDING

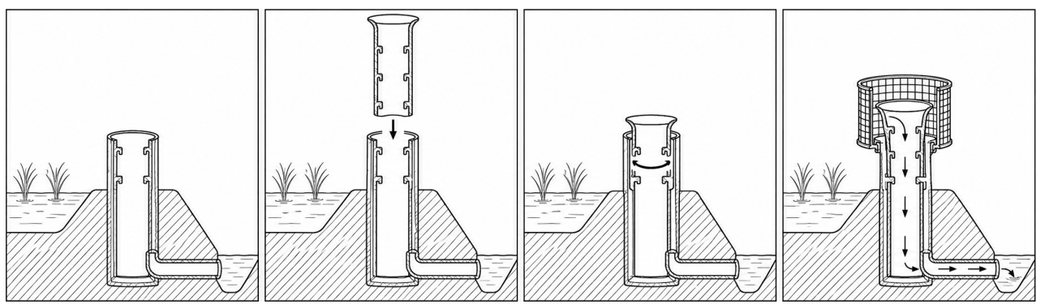
03 ORTHOGRAPHIC STUDIES / mm



FRONT / BODY **TOP / BORE** **SIDE / OUTLET**

Standpipe DIA 50; clear bore DIA 40; outlet DIA 30. Nominal overall H 240; screen OD and outlet projection TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 SET OUTER PIPE **2 INSERT RISER** **3 SET / LOCK** **4 FIT SCREEN**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-12 / P01	POKKALI-VENT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	12 / 28